

Day 1 Technical Task Record

Project Name

TicketOps API Bug Fixing (FastAPI)

Objective

Identify, debug, and fix issues in a FastAPI-based TicketOps API application and document the fixes.

Project Overview

The project consisted of a FastAPI application with multiple endpoints. The objective was to identify real bugs, correct them, verify the API functionality, and prepare the project for submission using Git and GitHub.

Environment & Tools

Python 3.12, FastAPI, Uvicorn, SQLite, VS Code, Jupyter, Git, GitHub.

Tasks Performed

1. Created a Python virtual environment.
2. Installed project dependencies from requirements.txt.
3. Resolved environment and activation issues.
4. Started the FastAPI application using Uvicorn.
5. Tested API endpoints using Swagger UI.
6. Fixed authentication-related issues.
7. Corrected API logic and validation problems.
8. Verified successful responses from all endpoints.
9. Documented the fixes in NOTES.md.
10. Uploaded the completed project to GitHub.

Key Files

app/main.py
app/auth.py
app/routers/items.py
app/routers/reports.py
requirements.txt
NOTES.md

Git Commands Used

git init
git add
git status
git commit
git remote add origin
git branch -M main
git push -u origin main

GitHub Repository

day1-broken-api

Key Learning Outcomes

- FastAPI project structure
- Debugging backend applications
- REST API testing using Swagger UI
- Authentication handling
- Git & GitHub workflow
- Technical documentation

Conclusion

Successfully fixed the TicketOps API issues, validated the endpoints, documented the work, and published the completed project to GitHub.